

Hypothesis Testing

Plan for today

- Testing logic: nulls, alternatives and evidence.
- Critical values, p -values and confidence intervals.
- Regression tests and the choice of standard errors.
- Type I and Type II errors; power and significance.
- Monte Carlo experiments, sample size and multiple testing.
- A preview of Wald, likelihood-ratio and score tests.

Appendix non-examinable

Inverting tests and pre-testing bias.

Testing and estimation

Estimation versus testing

Estimation

What is the value of the target?

Report a number such as $\hat{\beta}$.

Testing

Is a particular claim compatible
with the data?

Start from a null hypothesis H_0 .

Decide whether the evidence is
strong enough to reject it.

$$H_0 : \beta = \beta_0 \quad \text{versus} \quad H_1 : \beta \neq \beta_0.$$

The RCT: estimation and testing

Recall Lecture 1: we estimated the tutoring policy's effect using an RCT.

Estimation

What is our best guess for the effect of the policy?

$$\hat{\beta} = \bar{Y}_1 - \bar{Y}_0 = 9.33.$$

Testing

Do we have enough evidence to say that the policy had a positive effect?

Could sampling variation explain the difference if the policy had no effect?

A positive estimate **alone** does not establish a positive effect.

Null and alternative

- The null hypothesis is the claim we initially put on trial. For the opening examples, suppose $\beta_0 = 0$:

$$H_0 : \beta = 0.$$

- The alternative says what kind of departure would count as evidence against it.

$$H_1 : \beta > 0, \quad H_1 : \beta < 0, \quad H_1 : \beta \neq 0.$$

- The alternative determines which outcomes are called “extreme.”
- **These are just starting examples.** We can also test several coefficients jointly, nonlinear relationships, or claims about an entire distribution.

Are we asking whether H_0 is more likely than H_1 ?

No.

Comparing the probabilities of H_0 and H_1 given the data is a Bayesian testing question.

The best way to understand frequentist testing is through an analogy: a courtroom.

The legal analogy

- Start with innocent until proven guilty.
- Require evidence beyond reasonable doubt.
- Beyond this threshold: convict. Otherwise: do not convict.

Fingerprints on the murder weapon

If the person were innocent, how likely would we be to find their fingerprints on the knife?

A Bayesian question asks how likely guilt is *given the evidence*.
That also depends on the prior odds of guilt.

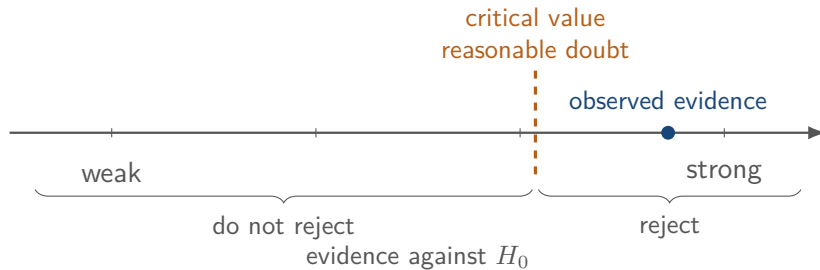
Assume the null

1. Assume H_0 is true.
How would the statistic behave in repeated samples?
2. Set a critical value c in advance.
This is our threshold for overturning H_0 .
3. Compare the observed statistic with the threshold.
Beyond it: reject H_0 . Otherwise: do not reject.

For a two-sided test of $H_0 : \beta = 0$,

$$|\hat{\beta}_{obs}| > c \implies \text{reject } H_0.$$

Evidence on a line



The critical value turns a continuous measure of evidence into a decision.

Proof by contradiction

Proof by contradiction

Assume A is true.

$A \Rightarrow B$.

B is false.

$\therefore A$ is false.

Hypothesis test

Assume H_0 is true.

$H_0 \Rightarrow \hat{\beta}$ has a known distribution.

$\hat{\beta}_{obs}$ is unlikely under this distribution

$\therefore H_0$ is probably not true.

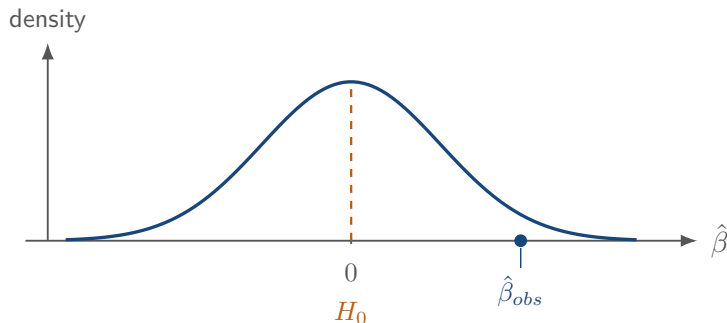
Why “reject” and “do not reject”?

- The test is designed to find evidence *against* H_0 .
- If the evidence is strong, we reject H_0 .
- If the evidence is not strong, we have not proved H_0 .
- We have only failed to find enough evidence to overturn it.

“Do not reject” means insufficient evidence against H_0 , **not proof that H_0 is true.**

Three equivalent languages

The null distribution



- For these opening examples, suppose $\beta_0 = 0$, so $H_0 : \beta = 0$.
- The **estimator** $\hat{\beta}$ is a random rule. The curve shows its distribution across repeated samples under H_0 .
- The **estimate** $\hat{\beta}_{obs}$ is the number from our one realized sample. We compare this point with the curve.

One-sided or two-sided?

Suppose we are testing $H_0 : \beta = 0$.

Two-sided

$$H_1 : \beta \neq 0$$

Reject if $\hat{\beta}_{obs}$ is
too high or too low.

A critical value in each tail.

One-sided

$$H_1 : \beta > 0$$

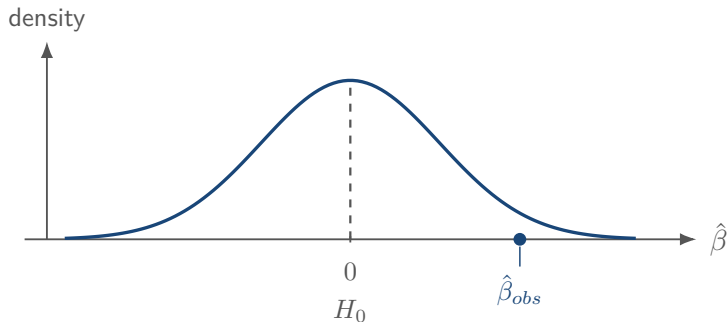
Reject if $\hat{\beta}_{obs}$ is
too high.

A critical value in one tail.

For example, a right-sided test may suit an assumption that β cannot be negative.

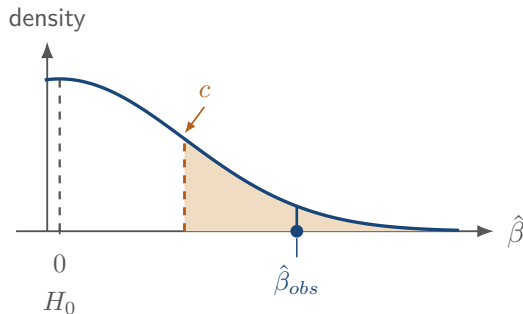
We could instead test $H_1 : \beta < 0$ (too low): choose the direction in advance.

One-sided test: the estimate



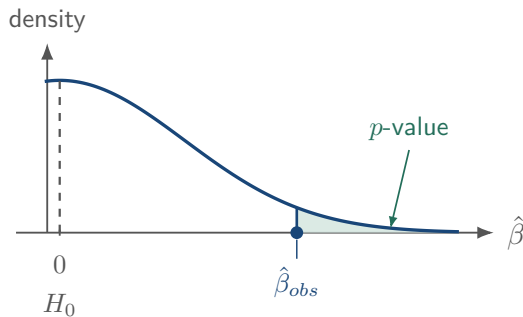
- Suppose the alternative is $H_1 : \beta > 0$.
- Large positive values of $\hat{\beta}$ are evidence against H_0 .
- We start with $\hat{\beta}$ directly. Tests usually use a **test statistic**; we will introduce that later.

One-sided test: critical value



- Reject H_0 if $\hat{\beta}_{obs} > c$ (here $\beta_0 = 0$).
- The orange area is the significance level: $\alpha = \mathbb{P}_{H_0}(\hat{\beta} > c)$.
- For this null distribution, choosing c is equivalent to choosing α .

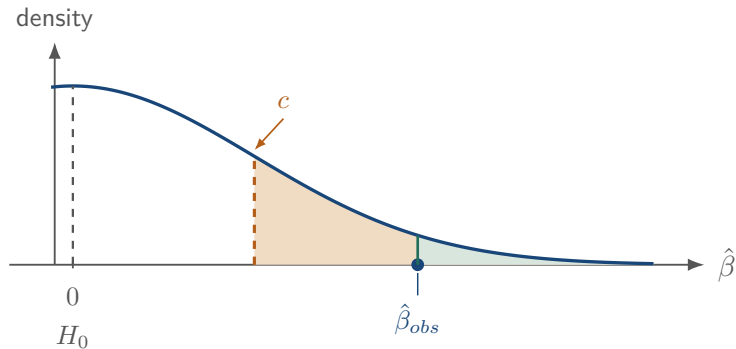
One-sided test: p -value



The green area is the p -value:

the probability, under H_0 , of an estimate at least this far to the right.

One-sided test: $\hat{\beta}_{obs} > c$

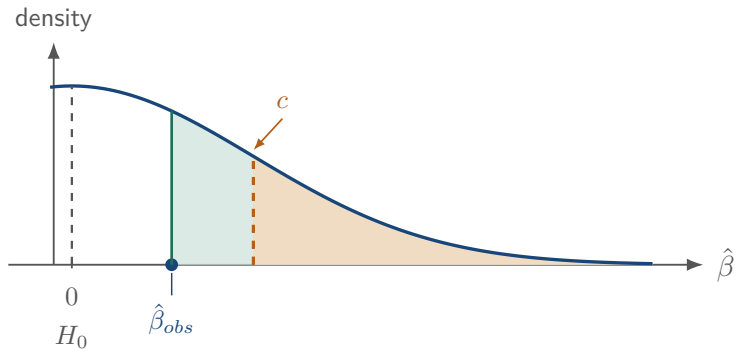


Green area = p -value (p).

Orange + green areas = significance level (α).

$$\hat{\beta}_{obs} > c \quad \text{and} \quad p < \alpha.$$

One-sided test: $\hat{\beta}_{obs} < c$

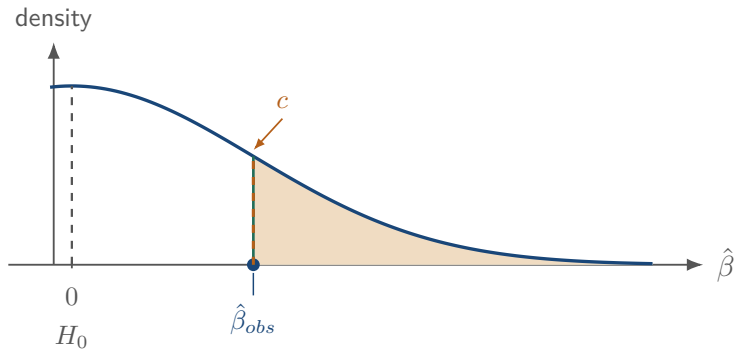


Green + orange areas = p -value (p).

Orange area = significance level (α).

$$\hat{\beta}_{obs} < c \quad \text{and} \quad p > \alpha.$$

One-sided test: $\hat{\beta}_{obs} = c$



Orange area = p -value (p) = significance level (α).

$$\hat{\beta}_{obs} = c \quad \text{and} \quad p = \alpha.$$

One-sided test: a general rule

$$H_0 : \beta = 0 \quad \text{versus} \quad H_1 : \beta > 0.$$

$$\text{reject } H_0 \iff \hat{\beta}_{obs} > c \iff p < \alpha.$$

What a p -value tells us

A p -value measures how extreme our observed estimate is *under* H_0 . For this right-sided test,

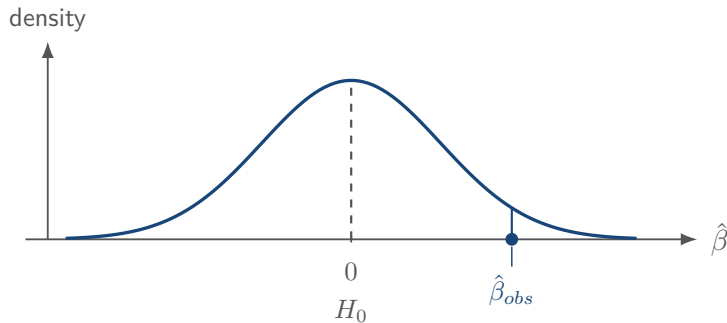
$$p = \mathbb{P}_{H_0}(\hat{\beta} \geq \hat{\beta}_{obs}).$$

- It is just *another way* to apply the same rejection rule:

$$p < \alpha \iff \hat{\beta}_{obs} > c.$$

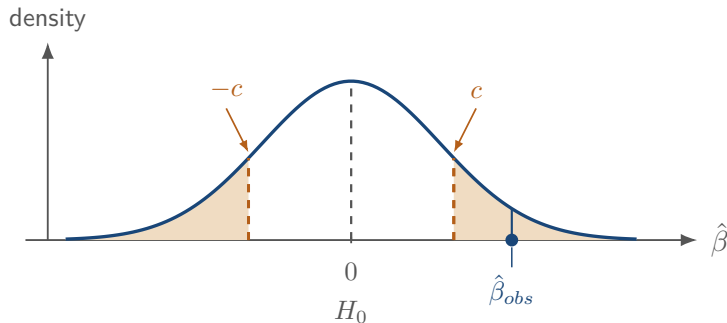
- *A common mistake:* p is *not* the probability that H_0 is true.
- That sort of thing occurs in Bayesian testing.

Two-sided test: the estimate



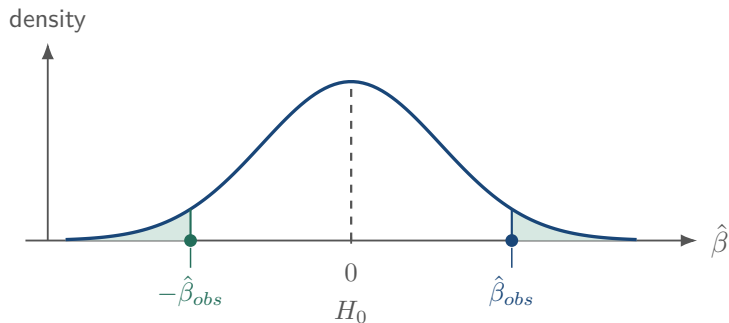
- Suppose the alternative is $H_1 : \beta \neq 0$.
- Large positive *or* negative estimates are evidence against H_0 .
- We compare $|\hat{\beta}_{obs}|$ with the cutoff: both directions count.

Two-sided test: critical values



- Reject H_0 if $|\hat{\beta}_{obs}| > c$ (here $\beta_0 = 0$).
- The two orange areas together give the significance level: $\alpha = \mathbb{P}_{H_0}(|\hat{\beta}| > c)$.
- Here the null distribution is symmetric: each tail has probability $\alpha/2$.

Two-sided test: p -value



The two green areas together give the p -value:

the null probability of an estimate at least this far from zero in either direction.

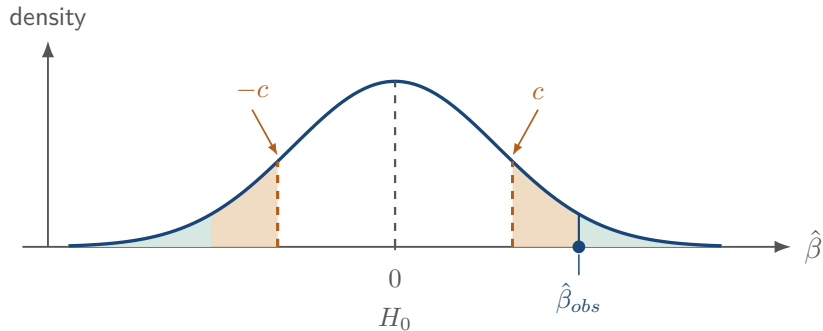
$$p = \mathbb{P}_{H_0}(|\hat{\beta}| \geq |\hat{\beta}_{obs}|).$$

Two-sided test: a general rule

$$H_0 : \beta = 0 \quad \text{versus} \quad H_1 : \beta \neq 0.$$

$$\text{reject } H_0 \iff |\hat{\beta}_{obs}| > c \iff p < \alpha.$$

Two-sided test: $\hat{\beta}_{obs} > c$

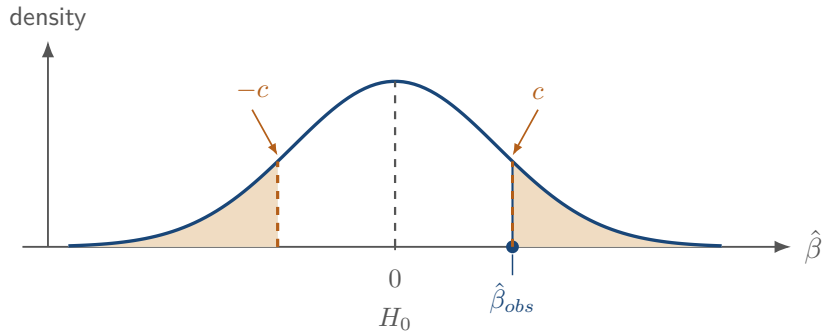


Green areas = p -value (p).

Orange + green areas = significance level (α).

$$\hat{\beta}_{obs} > c \quad \text{and} \quad p < \alpha.$$

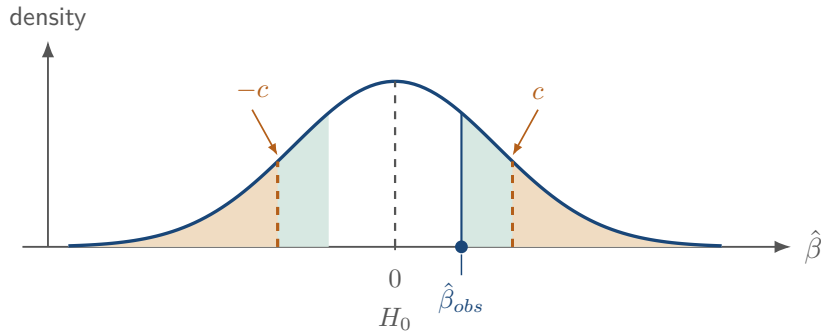
Two-sided test: $\hat{\beta}_{obs} = c$



Orange areas = p -value (p) = significance level (α).

$$\hat{\beta}_{obs} = c \quad \text{and} \quad p = \alpha.$$

Two-sided test: $0 \leq \hat{\beta}_{obs} < c$

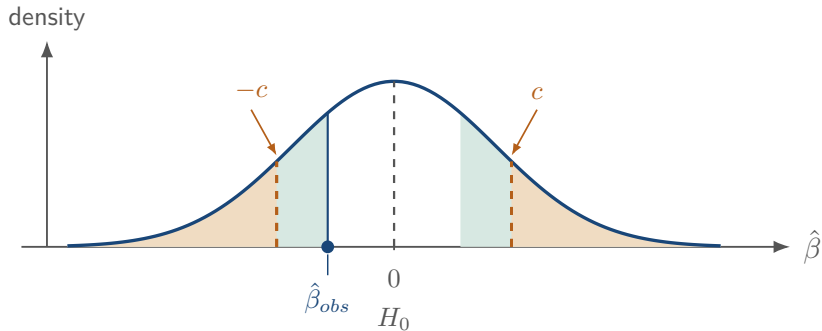


Green + orange areas = p -value (p).

Orange areas = significance level (α).

$$0 \leq \hat{\beta}_{obs} < c \quad \text{and} \quad p > \alpha.$$

Two-sided test: $-c < \hat{\beta}_{obs} < 0$

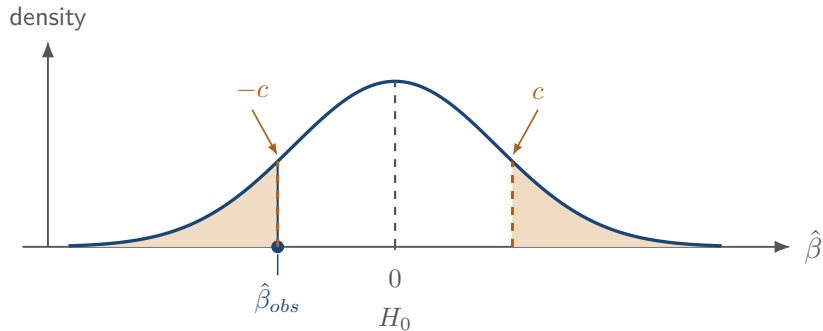


Green + orange areas = p -value (p).

Orange areas = significance level (α).

$$-c < \hat{\beta}_{obs} < 0 \quad \text{and} \quad p > \alpha.$$

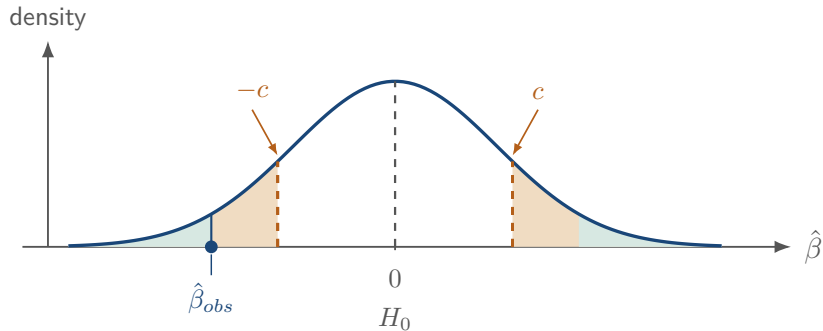
Two-sided test: $\hat{\beta}_{obs} = -c$



Orange areas = p -value (p) = significance level (α).

$$\hat{\beta}_{obs} = -c \quad \text{and} \quad p = \alpha.$$

Two-sided test: $\hat{\beta}_{obs} < -c$



Green areas = p -value (p).

Orange + green areas = significance level (α).

$$\hat{\beta}_{obs} < -c \quad \text{and} \quad p < \alpha.$$

Testing with confidence intervals

We want a third, equivalent rule:

$$\text{reject } H_0 \iff 0 \notin C_{1-\alpha}.$$

Here 0 is our hypothesized value.

You may have seen a 95% confidence interval like

$$C_{1-\alpha} = \left[\hat{\beta}_{obs} - 1.96 se, \hat{\beta}_{obs} + 1.96 se \right].$$

Set the formula aside: let us build the interval from **first principles**.

What does 95% confidence mean?

The interval is random. The true β is fixed.

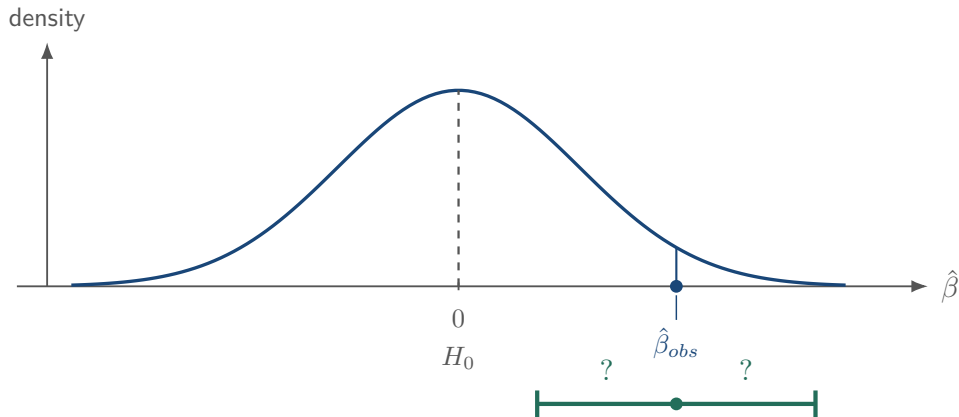
Different samples give different interval endpoints.

A 95% confidence procedure produces intervals that contain the true β in 95% of repeated samples.

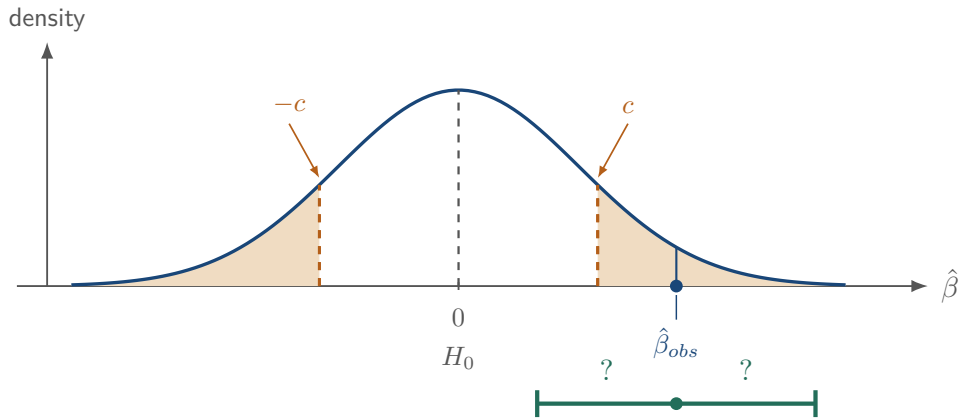
This is not a 95% belief about β given our data.

That would be a Bayesian statement.

How wide should the confidence interval be?

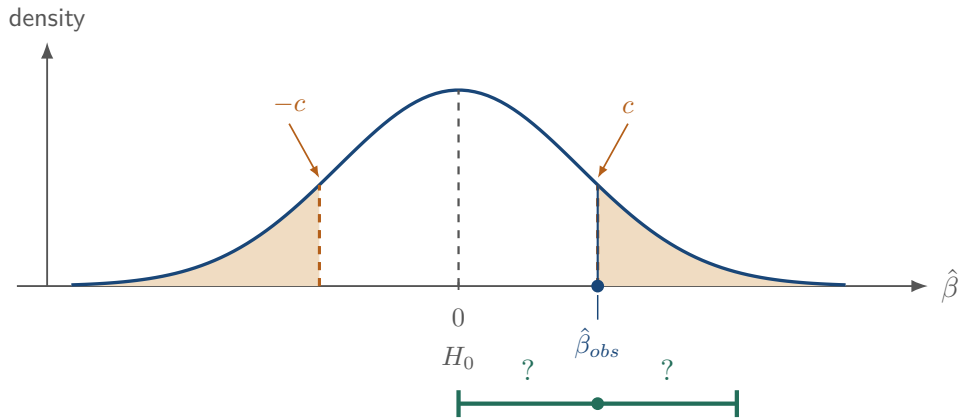


Intervals and critical values



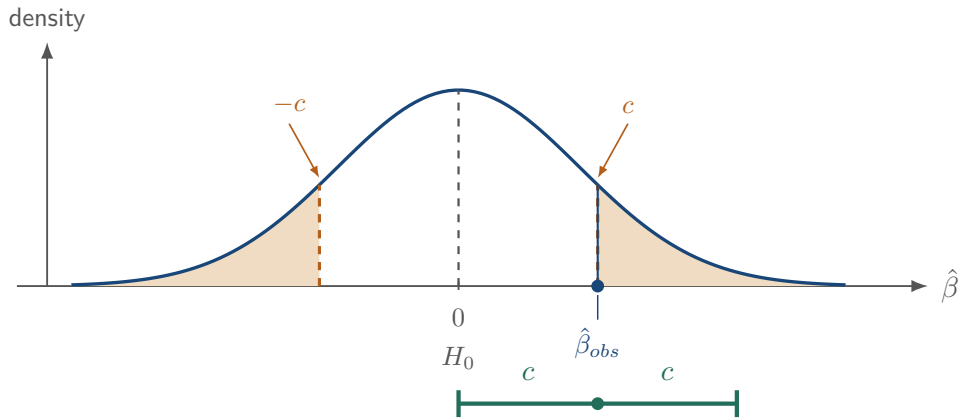
We want $0 \notin C_{1-\alpha} \iff |\hat{\beta}| > c.$

Confidence intervals: $\hat{\beta}_{obs} = c$



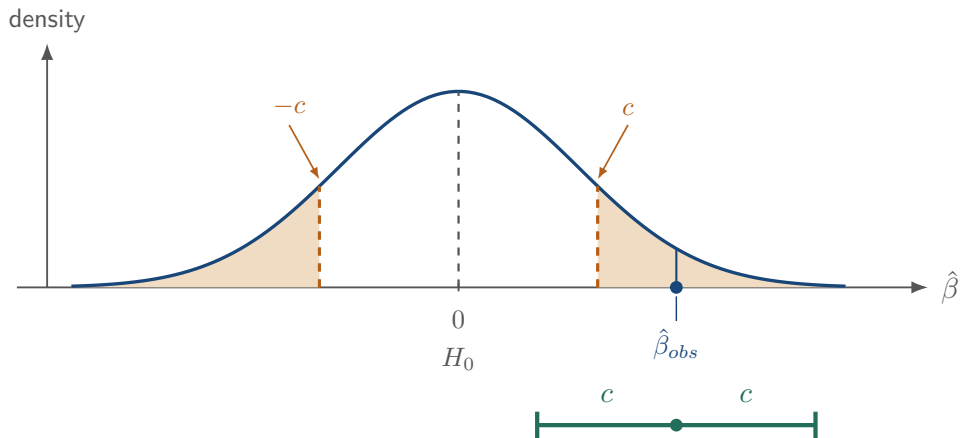
We want $0 \notin C_{1-\alpha} \iff |\hat{\beta}| > c.$

The solution



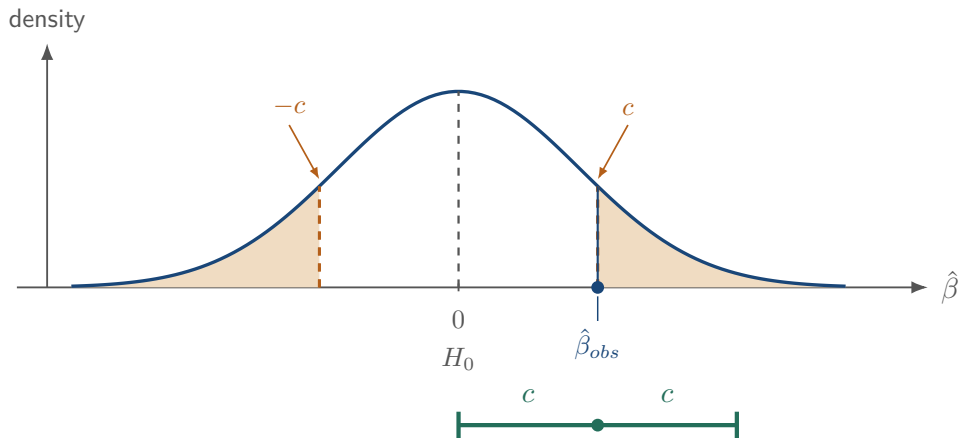
$$0 \notin C_{1-\alpha} \iff |\hat{\beta}| > c.$$

Confidence intervals: $\hat{\beta}_{obs} > c$



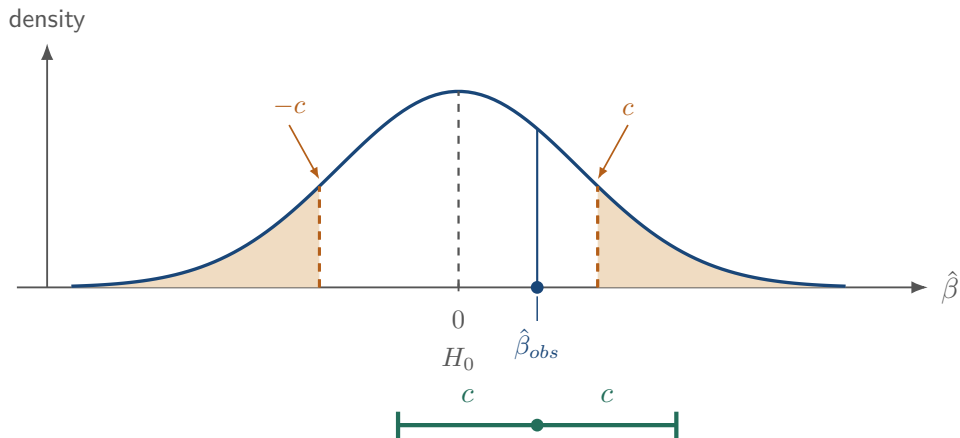
$$\hat{\beta}_{obs} > c \quad \text{and} \quad 0 \notin C_{1-\alpha}.$$

Confidence intervals: $\hat{\beta}_{obs} = c$



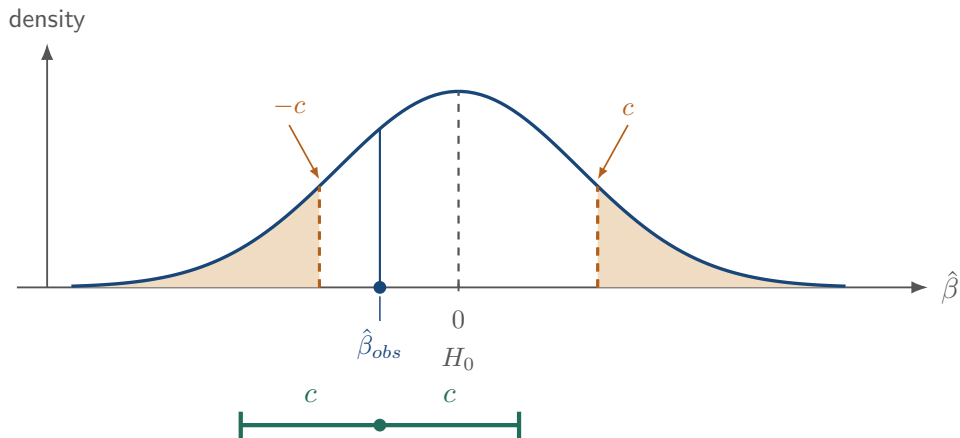
$\hat{\beta}_{obs} = c$, and zero lies on the boundary of $C_{1-\alpha}$.

Confidence intervals: $0 \leq \hat{\beta}_{obs} < c$



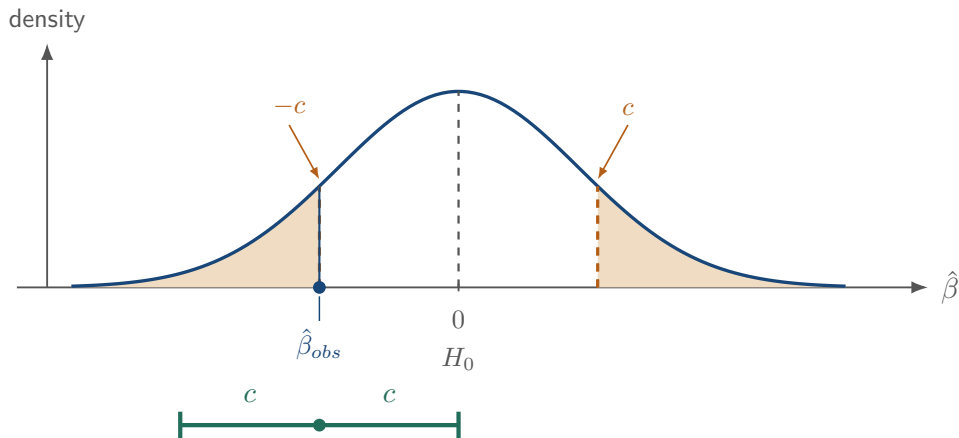
$$0 \leq \hat{\beta}_{obs} < c \quad \text{and} \quad 0 \in C_{1-\alpha}.$$

Confidence intervals: $-c < \hat{\beta}_{obs} < 0$



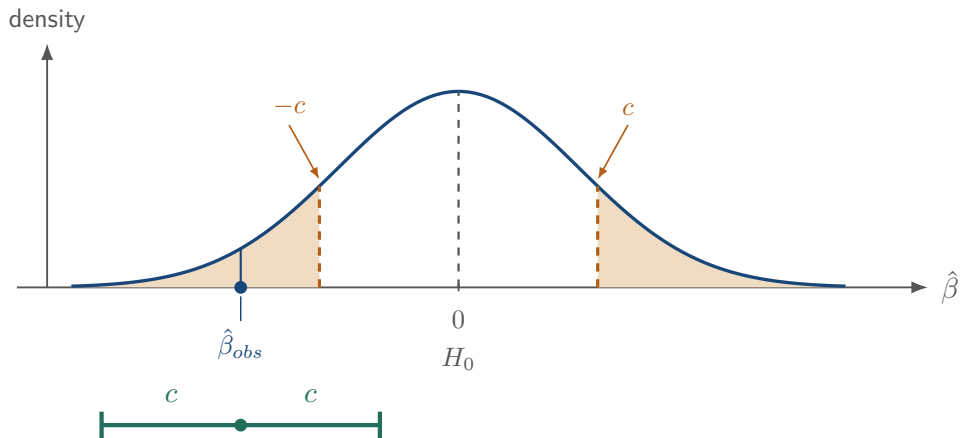
$$-c < \hat{\beta}_{obs} < 0 \quad \text{and} \quad 0 \in C_{1-\alpha}.$$

Confidence intervals: $\hat{\beta}_{obs} = -c$



$\hat{\beta}_{obs} = -c$, and zero lies on the boundary of $C_{1-\alpha}$.

Confidence intervals: $\hat{\beta}_{obs} < -c$



$$\hat{\beta}_{obs} < -c \quad \text{and} \quad 0 \notin C_{1-\alpha}.$$

Two-sided tests: three rules

$$H_0 : \beta = 0 \quad \text{versus} \quad H_1 : \beta \neq 0.$$

$$\text{reject } H_0 \iff |\hat{\beta}_{obs}| > c \iff p < \alpha \iff 0 \notin C_{1-\alpha}.$$

The $(1 - \alpha)$ confidence interval is

$$C_{1-\alpha} = \left[\hat{\beta}_{obs} - c, \hat{\beta}_{obs} + c \right].$$

From $\hat{\beta}$ to test statistics

What do we need to carry out a test?

To apply a rejection rule, we need to calculate both the **observed value** and the **critical value**.

But even under H_0 , the distribution of $\hat{\beta}$ depends on unknown V .
So $\hat{\beta}$ is not pivotal: we cannot obtain its cutoff directly.

We want a test statistic that is both:

1. **Pivotal**: its null distribution has no unknown parameters.
2. **Calculable**: we can compute its observed value from the data.

From $\hat{\beta}$ to Z

Under suitable assumptions and $H_0 : \beta = \beta_0$, with $\beta_0 \in \mathbb{R}$,

$$\hat{\beta} \sim N(\beta_0, V) \quad \text{under } H_0.$$

We can calculate $\hat{\beta}$, but it is **not pivotal**: V is unknown.

Standardising removes the unknown parameters from the distribution:

$$Z = \frac{\hat{\beta} - \beta_0}{\sqrt{V}} \sim N(0, 1) \quad \text{under } H_0.$$

Z is pivotal, but we **cannot calculate it**: we still need V .

The T statistic

For coefficient j , use the corresponding diagonal component of $s^2(X'X)^{-1}$:

$$\hat{V} = s^2[(X'X)^{-1}]_{jj},$$

where s^2 is an estimator of σ^2 .

$$T = \frac{\hat{\beta} - \beta_0}{\sqrt{\hat{V}}} = \frac{\hat{\beta} - \beta_0}{\sqrt{V}} \frac{\sqrt{V}}{\sqrt{\hat{V}}} = \frac{Z}{\sqrt{\hat{V}/V}}.$$

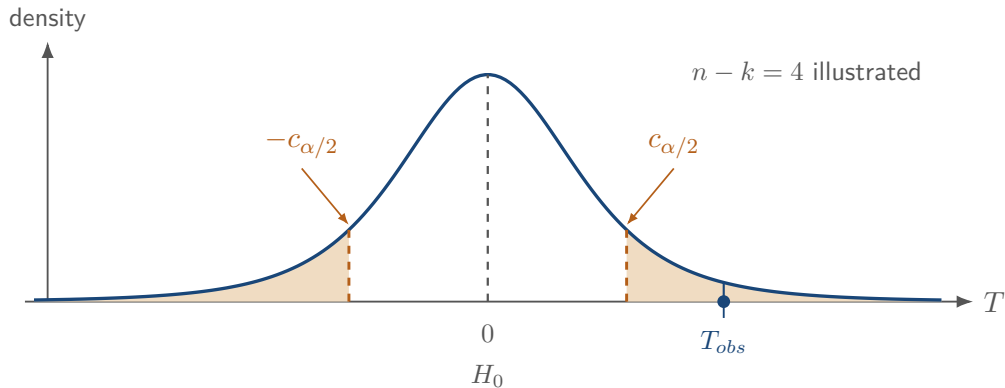
Under Gauss–Markov assumptions + normal errors,

$$Q = (n - k)\hat{V}/V \sim \chi_{n-k}^2, \quad Q \text{ independent of } Z \quad \text{under } H_0.$$

$$\text{Therefore} \quad T := \frac{\hat{\beta} - \beta_0}{\sqrt{\hat{V}}} = \frac{Z}{\sqrt{Q/(n - k)}} \sim t_{n-k} \quad \text{under } H_0.$$

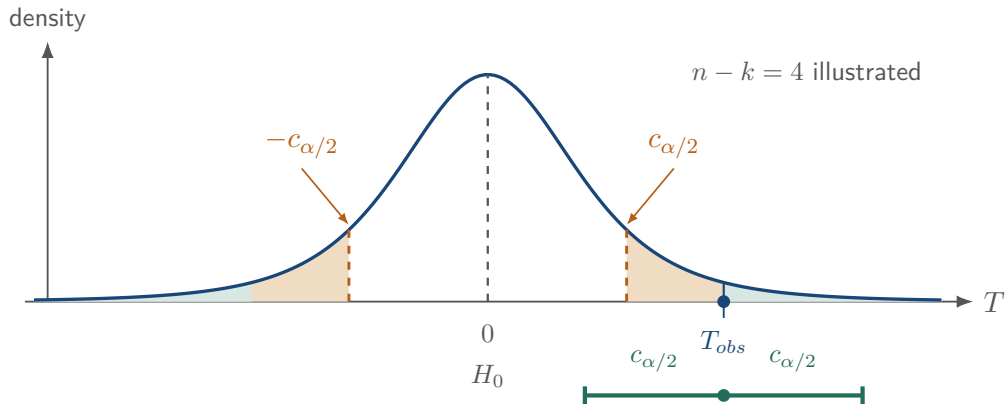
T is both **pivotal** and **calculable**.

Critical values for T



For $\alpha = 0.05$, we have $c_{\alpha/2} \approx 1.96$
when n is large.

T : p -values and intervals



Green areas = p -value (p).

Orange + green areas = significance level (α).

T tests: three rules

$$H_0 : \beta = \beta_0 \quad \text{versus} \quad H_1 : \beta \neq \beta_0.$$

$$\text{reject } H_0 \iff |T| > c_{\alpha/2} \iff p < \alpha \iff \beta_0 \notin C_{1-\alpha}.$$

Here $T = (\hat{\beta} - \beta_0)/se$, where $se > 0$ is the standard error.

$$\begin{aligned} 0 \in [T - c_{\alpha/2}, T + c_{\alpha/2}] &\iff -c_{\alpha/2} \leq T \leq c_{\alpha/2} \\ &\iff -c_{\alpha/2}se \leq \hat{\beta} - \beta_0 \leq c_{\alpha/2}se \\ &\iff \beta_0 \in \underbrace{[\hat{\beta} - c_{\alpha/2}se, \hat{\beta} + c_{\alpha/2}se]}_{C_{1-\alpha}}. \end{aligned}$$

So our earlier cutoff was $c = c_{\alpha/2}se$.

Significant, but important?

In a large sample, we estimate a £1 rise in weekly earnings.

Its standard error is £0.10.

Statistical significance

Against $H_0 : \beta = 0$,

$$T = 1/0.10 = 10.$$

Strong evidence of a nonzero effect.

Economic importance

Is £1 per week enough to matter?

That depends on the policy's costs and the economic context.

Report the **effect size and confidence interval**, as well as the p -value.

Summary

$$H_0 : \beta = \beta_0 \quad \text{versus} \quad H_1 : \beta \neq \beta_0.$$

Write $T_{obs} = (\hat{\beta} - \beta_0)/se$ and let $T \sim t_{n-k}$ under H_0 .

	Definition	Reject when
Critical value	$\Pr(T > c_{\alpha/2}) = \alpha/2$	$ T_{obs} > c_{\alpha/2}$
p -value	$p = \Pr(T \geq T_{obs})$	$p < \alpha$
Confidence interval	$C_{1-\alpha} = [\hat{\beta} - c_{\alpha/2}se, \hat{\beta} + c_{\alpha/2}se]$	$\beta_0 \notin C_{1-\alpha}$

Three ways of measuring the same thing.

They always give the same testing decision.

Reading Stata regression output non-examinable

$$y_i = \beta_0 + \beta_1 x_i + u_i : \quad H_0 : \beta_1 = 0 \quad \text{versus} \quad H_1 : \beta_1 \neq 0.$$

Choose $\alpha = 0.05$. Here $n = 8$, $k = 2$, so $n - k = 6$.

```
. regress y x
```

	y	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
	x	.7261905	.1860853	3.90	0.008	.2708561	1.181525
	_cons	.8571429	.9396847	0.91	0.397	-1.442183	3.156469

```
. display invttail(6, .025)  
2.4469118
```

Statistic vs cutoff

$$|T_{obs}| = 3.902 > 2.447$$

Two-sided p -value

$$p = 0.0080 < 0.05$$

95% CI for β_1

$$0 \notin [0.271, 1.182]$$

All three say: **reject H_0** .

Reading R regression output non-examinable

$$y_i = \beta_0 + \beta_1 x_i + u_i : \quad H_0 : \beta_1 = 0 \quad \text{versus} \quad H_1 : \beta_1 \neq 0.$$

Choose $\alpha = 0.05$. Here $n = 8$, $k = 2$, so $n - k = 6$.

```
> x <- 1:8; y <- c(2, 1, 4, 3, 6, 4, 7, 6)
> fit <- lm(y ~ x)
> summary(fit)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.8571      0.9397   0.912  0.39685
x              0.7262      0.1861   3.902  0.00796 **
> confint(fit, "x")
      2.5 %    97.5 %
x 0.2708561 1.181525
> qt(0.975, df.residual(fit))
[1] 2.446912
```

Statistic vs cutoff

$$|T_{obs}| = 3.902 > 2.447$$

Two-sided p -value

$$p = 0.0080 < 0.05$$

95% CI for β_1

$$0 \notin [0.271, 1.182]$$

All three say: **reject H_0** .

A Puzzle

non-examinable

A puzzle: one-sided intervals non-examinable

For a two-sided test, we have three equivalent rules:

$$\text{reject } H_0 \iff |T| > c_{\alpha/2} \iff p < \alpha \iff \beta_0 \notin C_{1-\alpha}.$$

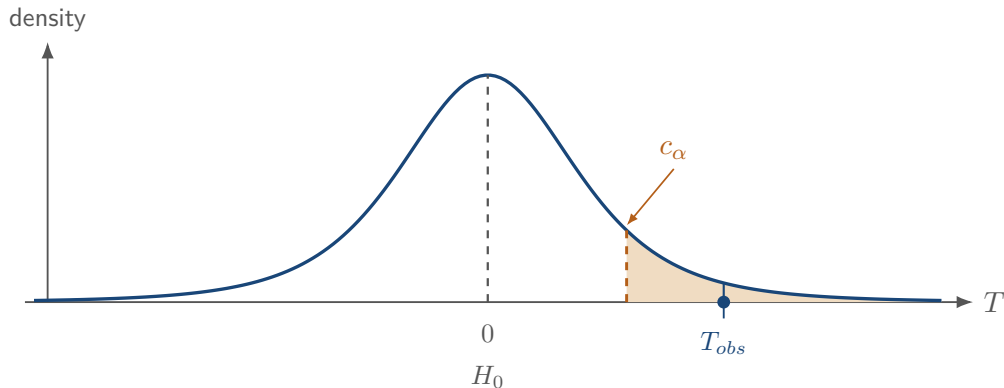
What about a **one-sided** test?

$$H_0 : \beta = \beta_0 \quad \text{versus} \quad H_1 : \beta > \beta_0.$$

What should the CI look like to give the **same decision**?

Finding a one-sided interval non-examinable

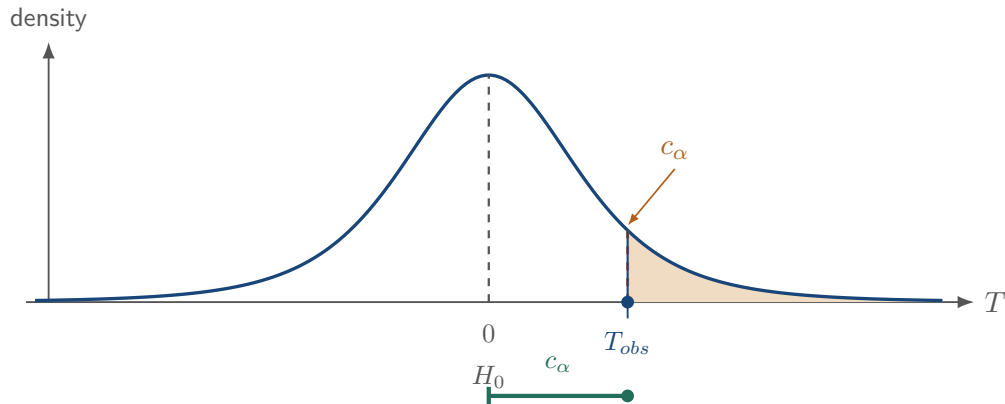
$$H_0 : \beta = \beta_0 \quad \text{versus} \quad H_1 : \beta > \beta_0.$$



$$0 \notin C_{1-\alpha} \iff T_{obs} > c_{\alpha}.$$

At the right-hand cutoff non-examinable

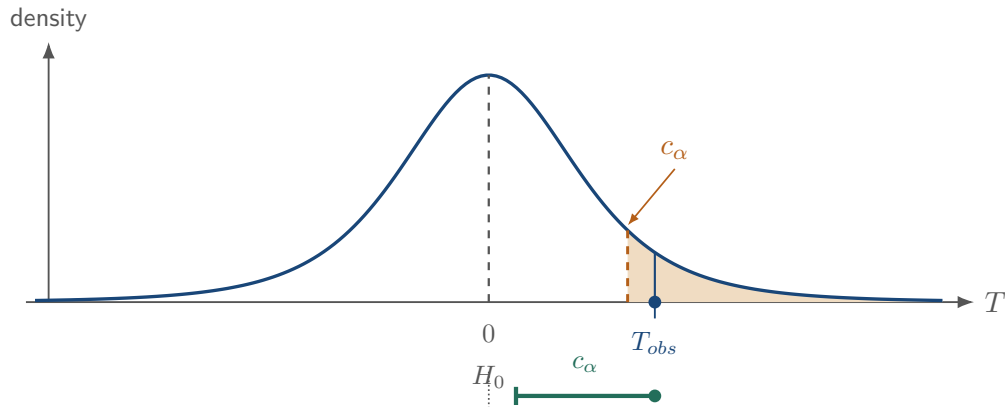
$$H_0 : \beta = \beta_0 \quad \text{versus} \quad H_1 : \beta > \beta_0.$$



At $T_{obs} = c_\alpha$, zero is on the boundary of $C_{1-\alpha}$.

Above the cutoff: $T_{obs} = c_\alpha + \epsilon$ non-examinable

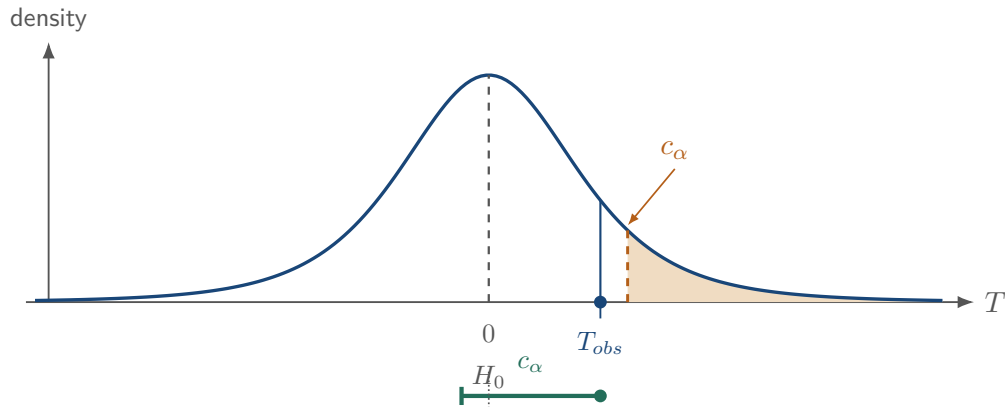
$$H_0 : \beta = \beta_0 \quad \text{versus} \quad H_1 : \beta > \beta_0.$$



$$T_{obs} = c_\alpha + \epsilon \implies 0 \notin C_{1-\alpha}.$$

Below the cutoff: $T_{obs} = c_\alpha - \epsilon$ non-examinable

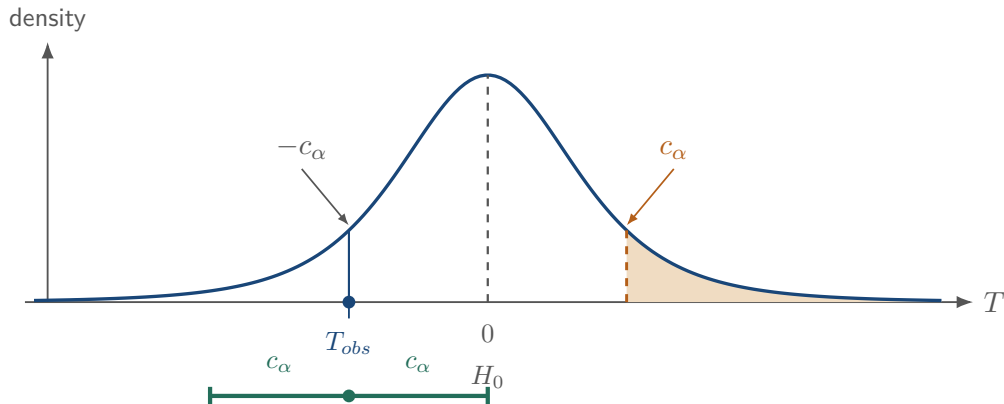
$$H_0 : \beta = \beta_0 \quad \text{versus} \quad H_1 : \beta > \beta_0.$$



$$T_{obs} = c_\alpha - \epsilon \implies 0 \in C_{1-\alpha}.$$

What if we used a symmetric interval? non-examinable

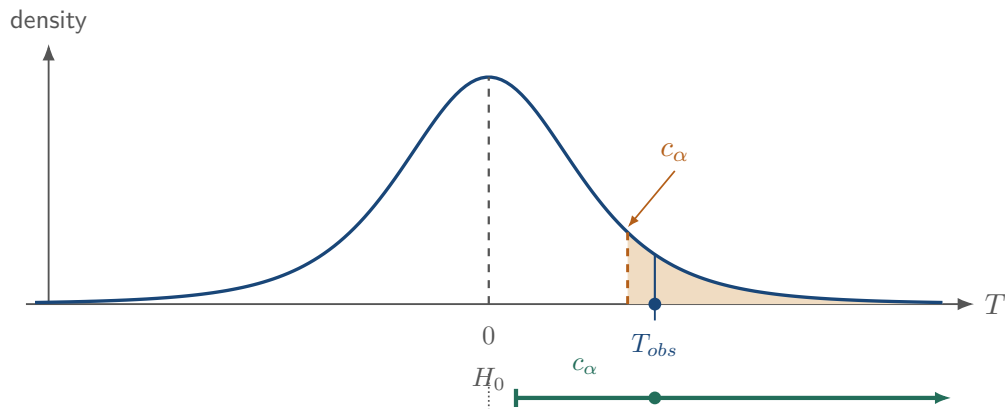
$$H_0 : \beta = \beta_0 \quad \text{versus} \quad H_1 : \beta > \beta_0.$$



At $T_{obs} = -c_\alpha$, zero is on the boundary.

But **we do not want this** for a one-sided test.

$$H_0 : \beta = \beta_0 \quad \text{versus} \quad H_1 : \beta > \beta_0.$$



$$C_{1-\alpha} = [T - c_\alpha, \infty).$$

One-sided tests: three rules non-examinable

$$H_0 : \beta = \beta_0 \quad \text{versus} \quad H_1 : \beta > \beta_0.$$

$$\text{reject } H_0 \iff T > c_\alpha \iff p < \alpha \iff \beta_0 \notin C_{1-\alpha}.$$

Here $T = (\hat{\beta} - \beta_0)/se$, where $se > 0$ is the standard error.

On the T scale, the confidence interval is $[T - c_\alpha, \infty)$.

$$0 \in [T - c_\alpha, \infty) \iff \beta_0 \in \underbrace{[\hat{\beta} - c_\alpha se, \infty)}_{C_{1-\alpha}}.$$

More complicated tests

Two complications

A preview of results developed in later courses.

Under Gauss–Markov assumptions + normal errors,

$$\hat{\beta} \sim N(\beta_0, V), \quad T \sim t_{n-k} \quad \text{under } H_0.$$

These are exact for $H_0 : \beta = \beta_0$ versus $H_1 : \beta \neq \beta_0$.

More complicated hypotheses

For example, testing a linear combination of coefficients.

Weaker assumptions

$\hat{\beta}$ need not be exactly normal;
 T need not be exactly t .

The asymptotic T test

Under suitable conditions, the **central limit theorem** gives

$$\hat{\beta} \overset{a}{\sim} N(\beta_0, V) \quad \text{under } H_0.$$

With a consistent variance estimator $\hat{V}$ and $se = \sqrt{\hat{V}}$,

$$T = \frac{\hat{\beta} - \beta_0}{se} \overset{a}{\sim} N(0, 1) \quad \text{under } H_0.$$

$\overset{a}{\sim}$ means **approximately distributed**, for large samples.

Two-sided 5% test: reject if $|T_{obs}| > 1.96$.

The p -value and CI rules apply approximately too.

Testing linear combinations

Test $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$, where $\theta = a'\beta$.

For example, $\theta = \beta_1 + \beta_2$.

$$\hat{\theta} = a'\hat{\beta}, \quad \text{Var}(\hat{\theta} \mid X) = a'Va.$$

Use $se(\hat{\theta}) = \sqrt{a'\hat{V}a}$, with $\hat{V}$ estimating $V = \text{Var}(\hat{\beta} \mid X)$.

$$T = \frac{\hat{\theta} - \theta_0}{\sqrt{a'\hat{V}a}} \stackrel{a}{\sim} N(0, 1) \quad \text{under } H_0.$$

Under suitable large-sample conditions.

Two-sided 5% test: reject if $|T_{obs}| > 1.96$.

Uncorrelated errors: choosing $\hat{V}$

$\Omega = \text{Var}(\varepsilon \mid X)$: the covariance matrix of the errors.

Homoscedastic

$$\Omega = \begin{pmatrix} \sigma^2 & 0 & 0 \\ 0 & \sigma^2 & 0 \\ 0 & 0 & \sigma^2 \end{pmatrix}$$

Conventional

$$\hat{V} = s^2(X'X)^{-1}$$

Heteroscedastic

$$\Omega = \begin{pmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & \sigma_3^2 \end{pmatrix}$$

White (Stata: `, robust`)

$$\hat{V}_{\text{HC}}$$

Standard errors are **estimated standard deviations**:

$$se(\hat{\beta}_j) = \sqrt{\hat{V}_{jj}}.$$

Dependent errors: choosing $\hat{V}$

$\Omega = \text{Var}(\varepsilon \mid X)$: the covariance matrix of the errors.

Clustered errors

$$\Omega = \begin{pmatrix} \sigma_1^2 & \omega_{12} & 0 \\ \omega_{12} & \sigma_2^2 & 0 \\ 0 & 0 & \sigma_3^2 \end{pmatrix}$$

Cluster-robust

$$\hat{V}_{\text{cluster}}$$

Serial correlation

$$\Omega = \begin{pmatrix} \sigma_1^2 & \omega_{12} & \omega_{13} \\ \omega_{12} & \sigma_2^2 & \omega_{23} \\ \omega_{13} & \omega_{23} & \sigma_3^2 \end{pmatrix}$$

HAC

$$\hat{V}_{\text{HAC}}$$

Standard errors are **estimated standard deviations**:

$$se(\hat{\beta}_j) = \sqrt{\hat{V}_{jj}}.$$

Errors and power

Decisions and reality

	H_0 true	H_0 false
Do not reject H_0	correct	Type II error
Reject H_0	Type I error	correct

The decision can be wrong in two different ways.

The two errors are not controlled in the same way.

Type I error

Reject H_0 when H_0 is true.

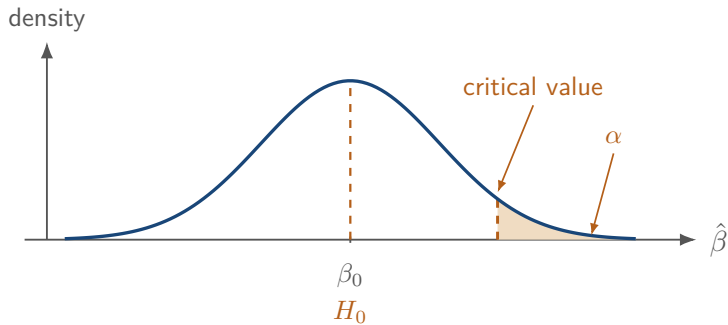
In the legal analogy: an innocent person goes to jail.

The significance level is

$$\alpha = \mathbb{P}(\text{reject } H_0 \mid H_0 \text{ true}).$$

We choose α directly by choosing the critical value.

Type I error as a tail area



- The rejection region has probability α when H_0 is true.

Type II error

Do not reject H_0 when H_0 is false.

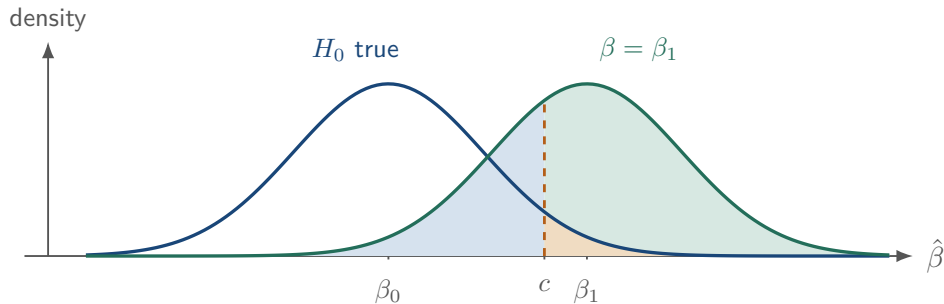
In the legal analogy: a guilty person goes free.

For a particular alternative value β_1 , write

$$1 - \text{Power}(\beta_1) = \mathbb{P}_{\beta_1}(\text{do not reject } H_0).$$

The Type II error depends on how far the truth is from the null.

Errors and power



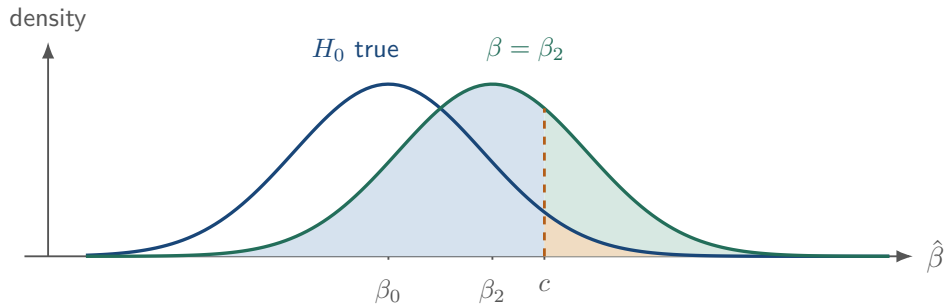
Orange area = Type I error probability, α .

Blue area = Type II error probability, $1 - \text{Power}(\beta_1)$.

Green + orange areas = $\text{Power}(\beta_1)$.

$$\text{Power}(\beta_1) = \mathbb{P}_{\beta_1}(\text{reject } H_0).$$

Power depends on the alternative



Orange area = Type I error probability, α .

Blue area = Type II error probability, $1 - \text{Power}(\beta_2)$.

Green + orange areas = $\text{Power}(\beta_2)$.

Power is a **function** of the alternative value of β .

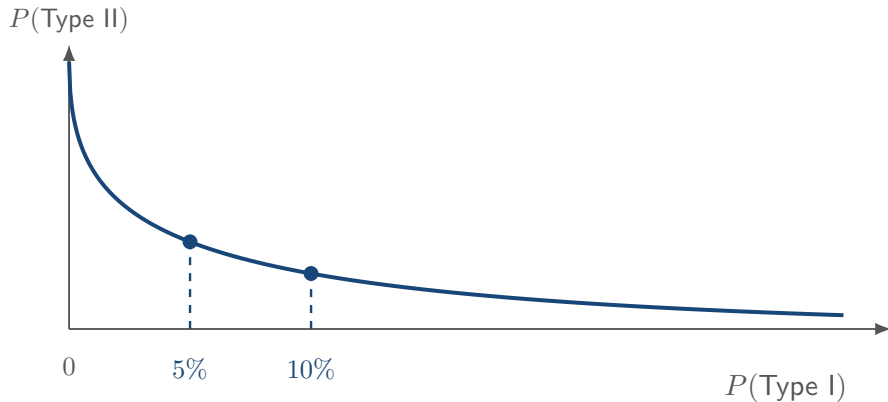
The legal tradeoff

- A stricter judge requires more evidence to send someone to jail.
- This reduces Type I errors: fewer innocent people are convicted.
- But it increases Type II errors: more guilty people go free.

There is a **tradeoff**: fewer Type I errors means more Type II errors.

The Type I–Type II tradeoff

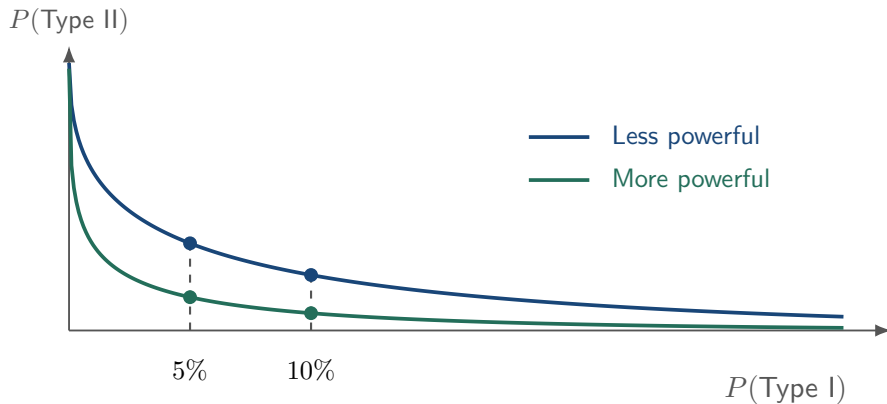
Hold the alternative value of β fixed.



Higher α means fewer Type II errors, but more Type I errors.

Improving power

Same alternative, same α : lower Type II error.



More data or a more efficient test can increase power.

Different names, same ideas

Language	Type I	Type II
Statistics	false rejection	false non-rejection
Machine learning	false positive	false negative
Signal detection	false alarm	miss
Medical screening	low specificity	low sensitivity
Legal analogy	innocent convicted	guilty acquitted

The labels change, but the tradeoff is the same.

Choosing α

Choose α **before looking at the estimate.**

Common choices are 10%, 5% and 1%.

Smaller α

Stronger evidence required to reject.

Fewer Type I errors, but lower power.

Prefer when false positives
are especially costly.

Larger α

Less evidence required to reject.

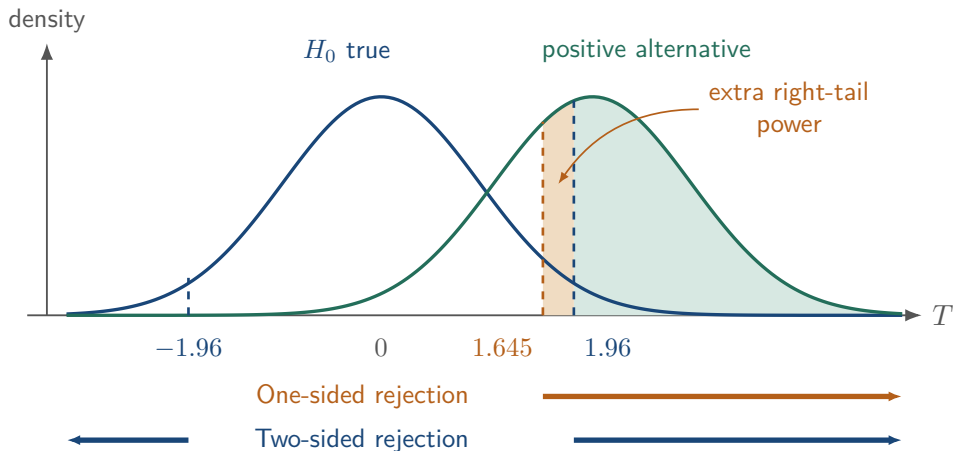
More Type I errors, but higher power.

Prefer when false negatives
are especially costly.

There is no universally correct α :
the choice reflects the **relative costs** of the two errors.

One-sided versus two-sided power

Both tests have $\alpha = 5\%$; the alternative is positive.



The price of one-sided power

A one-sided test puts all of α in the chosen tail.

The gain

More power in the
pre-specified direction.

The cost

Even extreme estimates in the
opposite direction do not reject.

Choose the direction from the substantive question,
before seeing the estimate.

Monte Carlo Experiments

What will these experiments show us?

We simulate many samples from a model where we know the truth.
Then we apply our tests and confidence intervals to each sample.

How big are **size, power and coverage** in practice?
What changes when we collect more data?

Minimum detectable effect

The smallest effect we can detect
with a chosen power, such as 80%,
for a given sample size and α .

Multiple testing

More tests mean more chances
for a false positive. **Bonferroni**
controls this risk across the tests.

A repeated-sampling model

Draw n independent observations from $N(\beta, \sigma^2)$, with σ known.

Estimate the mean

$$\hat{\beta} = \bar{Y}, \quad se = \sigma / \sqrt{n}$$

$$\hat{\beta} = \beta + se Z$$

Calculate the statistic

$$T = \frac{\hat{\beta} - \beta_0}{se} = \delta + Z$$

$$\delta = (\beta - \beta_0) / se$$

Here $Z \sim N(0, 1)$; δ measures the effect in standard-error units.

Our example: $n = 100$, $\sigma = 1$, $\beta_0 = 0$, $\alpha = 0.05$.

Under the alternative, $\beta = 0.20$.

Repeated samples

Draw n observations, calculate the test and CI, then repeat.

We use $R = 100,000$ repetitions.

Estimated size	Fraction rejected when H_0 is true
Estimated power	Fraction rejected at the chosen alternative
Estimated coverage	Fraction of CIs containing the true β

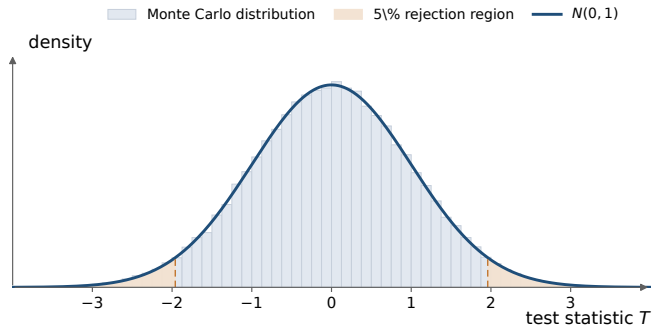
Larger n

More information per study;
usually greater power.

Larger R

More precise simulation results;
the test's power stays the same.

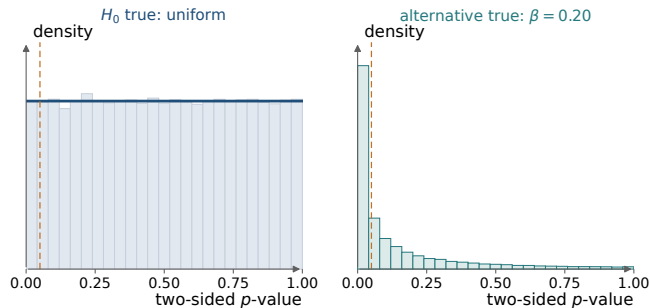
Simulating the null distribution



Under H_0 , $T \sim N(0, 1)$.

Simulated size: 4.9%. Target: 5%.

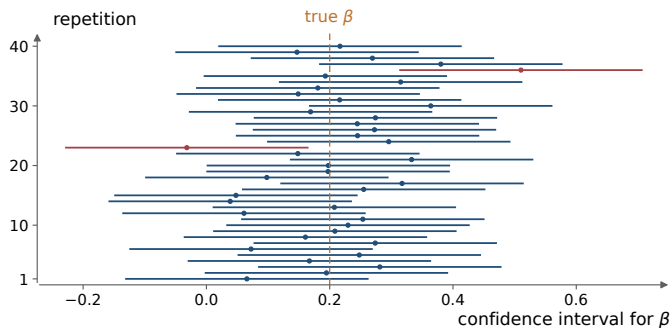
p -values in repeated samples



Under the null, $p \sim U[0, 1]$: $\Pr(p \leq a) = a$ in this exact model.

The share below α estimates size under H_0 (here 4.9%),
and power under the alternative.

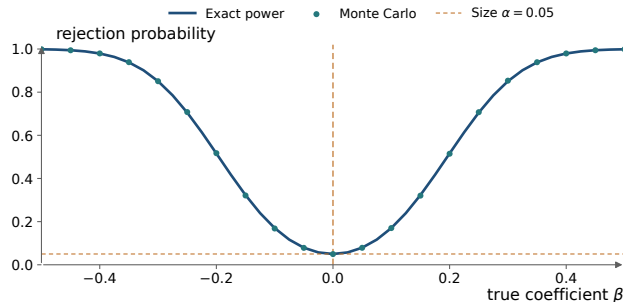
Simulating interval coverage



Each line is a new 95% interval. Red intervals miss the fixed β .

Across $R = 100,000$ samples, coverage is 95.0%.

Power is a function



For a two-sided test, with $\delta = (\beta - \beta_0)/se$,

$$\text{Power}(\beta) = 1 - \Phi(z_{1-\alpha/2} - \delta) + \Phi(-z_{1-\alpha/2} - \delta).$$

At β_0 , rejection probability is size; it rises in either direction.

Line: exact theory. Dots: simulation.

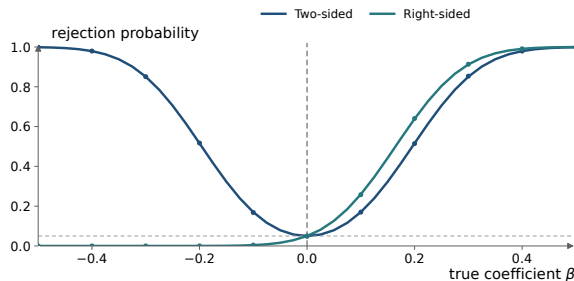
Direction changes power

More power in the
chosen direction.

At $\beta = 0.20$:

Right-sided: 64.1%

Two-sided: 51.8%



Right-sided power: $1 - \Phi(z_{1-\alpha} - \delta)$, where $\delta = (\beta - \beta_0)/se$.

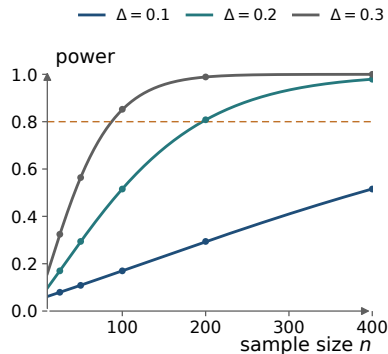
z_q is the q quantile of $N(0, 1)$.

Power and sample size

$$\delta = \frac{\sqrt{n}(\beta - \beta_0)}{\sigma}$$

More data or less noise
means **greater power**.

For a fixed positive effect,
right-sided power tends to 1
as $n \rightarrow \infty$.



An effect of $\Delta = 0.20$ needs $n = 197$ for two-sided 80% power.

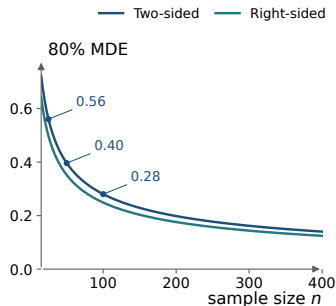
Minimum detectable effects

For two-sided 80% power, use the approximations

$$\text{MDE}_{0.8} \approx (z_{1-\alpha/2} + z_{0.8}) \frac{\sigma}{\sqrt{n}}$$

$$n \approx \left[\frac{(z_{1-\alpha/2} + z_{0.8})\sigma}{\Delta} \right]^2$$

Round the required n up.



Right-sided test: use $z_{1-\alpha}$ instead of $z_{1-\alpha/2}$.

Quadrupling n halves the standard error and the MDE.

Bonferroni: proof

Test m hypotheses, each at level α/m .

Let A_j be the event of a false rejection in test j .

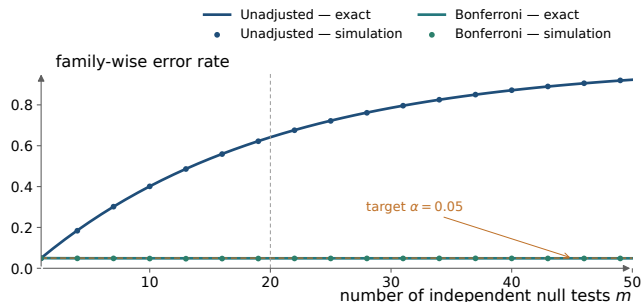
Then $\Pr(A_j) \leq \alpha/m$; the event is empty if that null is false.

$$\underbrace{\Pr\left(\bigcup_{j=1}^m A_j\right)}_{\text{family-wise error rate}} \stackrel{\text{union bound}}{\leq} \sum_{j=1}^m \Pr(A_j) \leq m \frac{\alpha}{m} = \alpha.$$

No independence assumption is needed.

The cost: smaller test levels mean less power; the bound can be conservative.

Monte Carlo: multiple-testing risk



With m independent true nulls and exact individual test sizes:

Unadjusted: $1 - (1 - \alpha)^m$. Bonferroni: $1 - (1 - \alpha/m)^m \leq \alpha$.

For $m = 20$: simulated 64.1% versus 4.8%
(exact: 64.2% versus 4.9%).

Bonferroni for 20 outcomes

Twenty independent true nulls, each tested at 5%.

At least one false rejection $1 - 0.95^{20} \approx 64.2\%$

Expected false rejections $20(0.05) = 1$

Bonferroni threshold $0.05/20 = 0.0025$

A smallest p -value of 0.018 is **not significant** after correction.

Pre-specify a primary outcome; do not select it by its p -value.

Multiple-outcome claims still need a multiplicity adjustment.

Summary

Wald, likelihood ratio and score

These three approaches may come up in other courses.

Wald	How far is the unrestricted estimate from the null?
Likelihood ratio	How much fit do we lose by imposing the null?
Score / LM	At the null, how strongly does the likelihood slope point away from it?

Different measures of evidence, the same testing logic:
calibrate under H_0 , then reject if the evidence is sufficiently unusual.

Key takeaways

Start from H_0 . Reject when the evidence is unusual under it.

Critical values, p -values and matching CIs give the same decision.

α controls Type I error; power measures our ability to detect an effect.

Probabilities describe repeated sampling, not whether H_0 is true.

More data can raise power. More tests require a multiple-testing adjustment.

Next time

Lecture 5: OLS and Asymptotics

Appendix

Inverting a test

non-examinable

Reinterpreting the T test non-examinable

Start with our familiar two-sided T test.

We will interpret the same rejection region using a horizontal density cutoff.

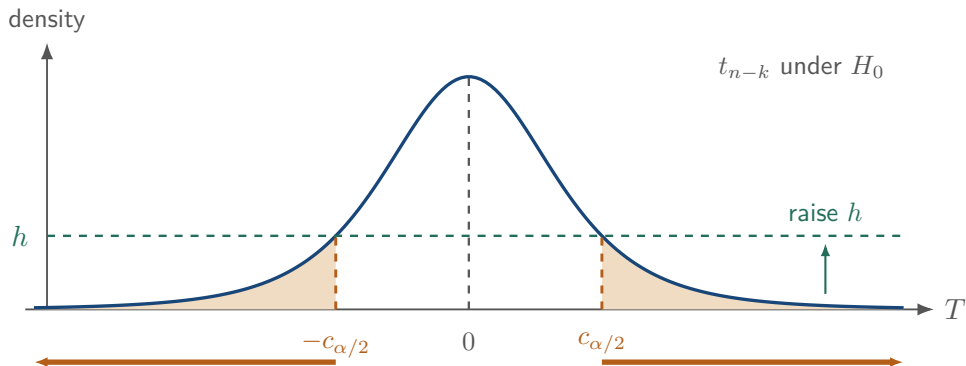
This gives us a way to think beyond symmetric distributions:
for example, χ^2 , F , or even a distribution with two peaks.

Inverting the test: keep every value of b that we do not reject.

The resulting confidence set / critical region not confidence interval.

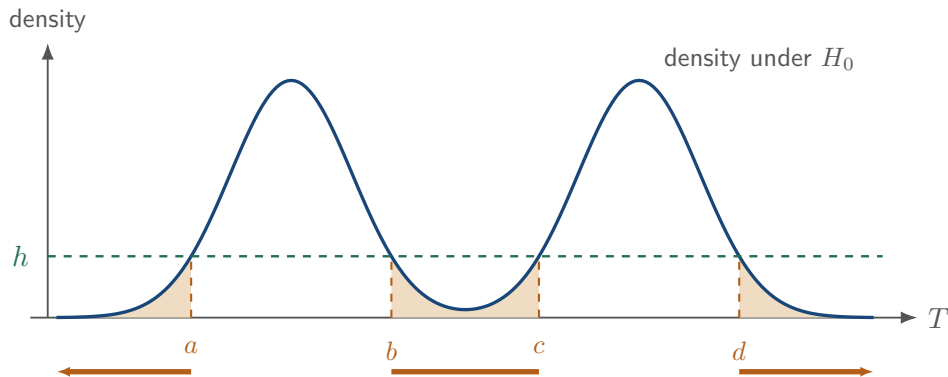
The T test: a horizontal slice non-examinable

The familiar two tails are exactly the values with **low density** under H_0 .



Raise h until the **two shaded tails** have total probability α .

Bimodality: disconnected rejection regions non-examinable



$R = (-\infty, a) \cup (b, c) \cup (d, \infty)$, with total probability $\alpha = 0.10$.

Reject H_0 if and only if $T_{obs} \in R$

Choosing the model first non-examinable

Suppose we want to test $H_0 : \beta_1 = 0$ in

$$y_i = \beta_0 + \beta_1 x_i + \gamma z_i + u_i.$$

A tempting procedure:

Test $\gamma = 0$; drop z if it is not significant.

Then test $\beta_1 = 0$ using the chosen model and the **same data**.

The final model was **selected by a preliminary test**.

Ordinary inference treats that model as if it had been fixed in advance.

Even when $\beta_1 = \gamma = 0$, the two statistics can be correlated.

In a normal approximation, with $\rho \neq 0$,

$$\begin{pmatrix} T_\beta \\ T_\gamma \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right).$$

Keep z when $S = \{|T_\gamma| > c\}$. We can write

$$T_\beta = \rho T_\gamma + \sqrt{1 - \rho^2} Z, \quad Z \sim N(0, 1), \quad Z \perp T_\gamma.$$

Because $\mathbb{E}[T_\gamma^2 \mid S] > 1$,

$$\mathbb{E}[T_\beta^2 \mid S] = 1 + \rho^2(\mathbb{E}[T_\gamma^2 \mid S] - 1) > 1.$$

So $T_\beta \mid S$ is **not standard normal**.

The usual critical value ignores selection: the final test's size is not guaranteed.

Pre-testing bias: the intuition non-examinable

Imagine repeating the whole procedure on many samples.

We keep z only when T_γ looks unusually large in absolute value.

Even if $\gamma = 0$, some samples get there by chance.

If T_β and T_γ are correlated, those selected samples also tend to have more extreme values of T_β .

The usual cutoff was calibrated for all samples.

We are now applying it within a selected subset.

The preliminary test is part of the procedure:
we cannot ignore it when calculating error probabilities.